The Perks of Being a First Mover

Does White's Advantage in Chess Change with Rating, Opening and Time Control?

CS215 Open Project · Final Report · Group 02

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https://github.com/cs215-cse-iitb/dai-aug-2026-openproject-group-2

Section 1: Meta Information

1. Open Project Title	The Perks of Being a First Mover: Does White's Advantage in Chess Change with Rating, Opening and Time Control?
2. Group ID	"02"
3. Mentor Name	Namish Bansal
4. Group Members and Roll Numbers	Sanidhya (25b0944); Daksh Maahor (25b0974); Sahil (25b0991); VadaPavJai (25b1030)
5. Honor Code and Discussions	None
6. LLM Usage Links	**[Readable links to all chat sessions / prompts used. If an AI agent was used, export its report as PDF, host it and link it here.]**
7. Video Presentation	**[link to video]**
8. GitHub Repository	https://github.com/cs215-cse-iitb/dai-aug-2026-openproject-group-2
9. GitHub Handles and Initial Commit	
Sanidhya	https://github.com/SS3642 earliest commit: October 21, 2026
Daksh Maahor	https://github.com/Daksh-Maahor earliest commit: November 3, 2026
Sahil	https://github.com/Sahil-Deo-IITB earliest commit: November 3, 2026
VadaPavJai	https://github.com/nauseated-potato earliest commit: October 18, 2026
10. Contribution Breakdown	Unequal

Work split (for reference).	Sanidhya	Statistical modelling: fitting h, confidence intervals and the z-tests.
	Daksh Maahor	Data cleaning and opening classification; win/draw/loss rates.
	Sahil	Data engineering: the C++ streaming parser and the Colab pipeline
	VadaPavJai	Visualisation: figures, the Plotly dashboard and the report.

Section 2: Technical Content

Core Problem Statement

We measure the first-move advantage on Lichess. Instead of raw win rates, we fit White's advantage as a number of *rating points*, h, after correcting each game for the rating gap between the players. Under the Elo model a player d points above the opponent scores $E(d) = 1/(1 + 10^{-d/400})$; we give White an extra h points, $E_i(h) = 1/(1 + 10^{-(R_i^{\mathrm{W}} + h - R_i^{\mathrm{B}})/400})$, and choose the h that makes expected scores equal actual scores ($S_i = 1, \frac{1}{2}, 0$). This is the maximum-likelihood estimate (Newton's method; standard error from the Fisher information). Overall, $h =$ **[HAll]** $\pm$ **[HAllCI]** rating points. We test three hypotheses:

- **H1 (Rating).** h peaks at intermediate ratings and shrinks at 2200+.
- **H2 (Openings).** Four aggressive 1.e4 gambits (King's, Evans, Danish, Smith-Morra) carry a larger h than four drawish openings (Petrov, Berlin, QGD, French Exchange), and the drawish ones have more draws at every rating.
- **H3 (Time control).** h grows from Bullet to Classical.

Data Source(s)

Lichess Open Database (`https://database.lichess.org`, CC0), standard rated games for **[MonthList]**: one zstd-compressed PGN per month, tens of GB each, openly downloadable. We use seven of 17 header tags: `Result`, `WhiteElo`, `BlackElo`, `Event`, `TimeControl`, `Termination`, `Opening`. We read **[GamesRead]** games and keep **[GamesUsed]** (**[PctUsed]**).

Month	Games read	Games kept	Kept
[filled in by white_advantage]			
Total	**[GamesRead]**	**[GamesUsed]**	**[PctUsed]**

Tools and Software Packages

A C++ program streams the zstd files, reads only header lines, and counts games by opening, speed, 100-point rating bracket and rating gap (two months on two threads, run on Google Colab, streaming directly from Lichess). It writes `counts.csv` (a few MB); all tables, tests, figures, this report (LaTeX with pgfplots) and an interactive Plotly dashboard (`output/final/dashboard.html`) are rebuilt from it in seconds. **[list exact Python/R packages and versions used for the analysis, if any]**

Challenges and Results Achieved

Cleaning and design. We drop games with no result (**[SkipNoResult]**), abandoned games (**[SkipAbandoned]**), missing ratings (**[SkipNoRating]**) and rating gaps above 100 (**[SkipGap]**). Games go into 100-point brackets by average rating and into a speed from `Event`. Openings are matched by Lichess name prefix (excluding gambit side-lines inside the drawish openings) and cover **[OpeningGames]** games (**[OpeningShare]**). With millions of games every difference has a tiny p-value, so we report effect sizes with 95% CIs and treat a few rating points as small. **[technical hurdles: e.g. file size/streaming, opening-name matching, memory, runtime]**

Overall. White wins **[WhiteAll]**, draws **[DrawAll]**, loses **[BlackAll]**; average score **[ScoreAll]** versus **[ExpectedAll]** expected from ratings alone, i.e. $h =$ **[HAll]** $\pm$ **[HAllCI]**.

H1: rating. **[is H1 supported? Describe the shape of h against rating]**

H2: openings. Each opening is fitted separately; groups are compared with a z-test, $\mathrm{SE}(\hat{h}_A - \hat{h}_D) =$

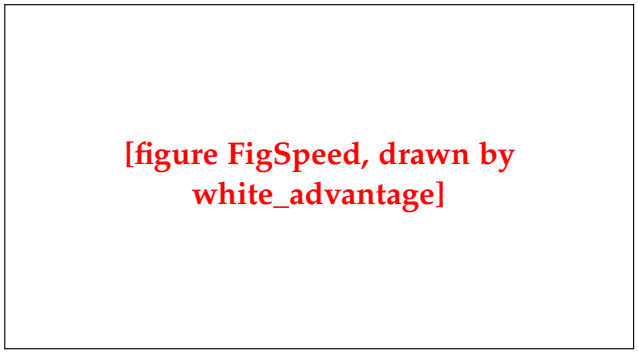


Figure 1: h per 100-point bracket, by speed (95% CIs; <1,000 games omitted).



Figure 2: h by rating: aggressive vs drawish openings.

$\sqrt{\mathrm{SE}_A^2 + \mathrm{SE}_D^2}$. Aggressive minus drawish: **[CompareDiff]** points ($z =$ **[CompareZ]**, $p =$ **[CompareP]**). **[is H2 supported?]**

Table 1: White's advantage by opening (all speeds).

Opening	Games	Avg rating	White	Draw	Black	h (pts)
[filled in by white_advantage]						
[filled in by white_advantage]						

H3: time control. Each speed is its own set of games with its own h; each speed is compared with the next faster one by the same z-test. **[is H3 supported?]**

Table 2: White's advantage by speed (all openings).

Speed	Games	Avg rating	White	Draw	Black	h (pts)
[filled in by white_advantage]						



Figure 3: h per opening with 95% CIs.



Figure 4: h per speed with 95% CIs.

Limitations. Opening choice is not random (association, not causation); the two opening groups have different players (see Fig. 8); ratings are noisy and Lichess is not FIDE; games are not independent (active players contribute thousands), so true uncertainty is somewhat larger than our CIs; one site and two months only.

Conclusion. **[answer the three questions in a few sentences]**

Section 3: Frequently Answered Questions (FAQ)

Q1. How many games did you analyse and where did they come from?
[GamesUsed] standard rated Lichess games from **[MonthList]** (**[GamesRead]** read, **[PctUsed]** kept), from the CC0 Lichess Open Database.

Q2. What is h and what is its overall value?
h is the number of Elo rating points that, added to White's rating, makes the Elo-expected scores match the observed scores. Overall $h =$ **[HAll]** $\pm$ **[HAllCI]**.

Q3. Which games did you exclude and why?
Games with no result, abandoned games, missing ratings and rating gaps above 100 points: very lopsided games say little about the first move.

Q4. Why rating points instead of raw win rates?
A raw win rate mixes the first move with the rating gap and the draw rate. Rating points correct each game for the gap, so groups with different rating mixes are comparable, and draws count as half a point.

Q5. Why not fit a single slope of h against rating?
The shape is not close to a straight line, so a slope would hide it; we fit h per 100-point bracket and read the shape from the plot.

Q6. With millions of games, isn't everything "significant"?
Yes, so we report effect sizes with 95% CIs and treat differences of a few rating points as small whatever the p-value.

Q7. Which openings were compared?
Aggressive: King's, Evans, Danish and Smith-Morra Gambits. Drawish: Petrov, Berlin, Queen's Gambit Declined, French Exchange (gambit side-lines excluded).

Q8. Does the data support the hypotheses?
[H1, H2, H3 each: supported / partly / not supported, with the key number]

Q9. Does the opening *cause* the difference in h?
Not necessarily. Players choose openings, and styles, preparation and ratings differ between groups, so our results show association only.

Q10. What would you do with more time?
[e.g. more months, cluster-robust CIs by player, adjusting for player ratings, FIDE/OTB data]

Section 4: Reflection Essay (Optional)

[Write your reflection here. Suggested prompts: key takeaways and skills learned; how the project went beyond textbook material; why real-world problems matter in a curriculum; relevance to placements and careers; navigating an open-ended, unguided problem; effort compared with standard homework; personal anecdotes, challenges and final thoughts.]

Summary

We measure the first-move advantage on Lichess using **[GamesUsed]** standard rated games from **[MonthList]**. Instead of comparing raw win rates, we fit White's advantage as a number of *rating points*, h, after correcting every game for the rating gap between the players. Overall White is worth $h =$ **[HAll]** $\pm$ **[HAllCI]** rating points. We then ask three questions: how h changes with playing strength, with the time control (Bullet to Classical), and with the opening, where we compare four openings with an aggressive reputation (all 1.e4 gambits) against four with a drawish reputation. **[one or two sentences with the main findings, written once the real results are in]**

1. Question and hypotheses

Chess is a game of perfect information in which White moves first. Database statistics have long shown that White scores a little better than Black, but a raw win rate mixes three things: the first move itself, the rating gap between the players, and how often games are drawn. We separate them and test:

H1 **Rating (from our proposal).** White's advantage is not constant across skill levels. It peaks at intermediate ratings, where players can exploit the initiative but defend poorly, and shrinks at 2200+, where Black knows how to neutralise it.

H2 **Openings (new for the final stage).** Aggressive openings, where White gives up material for the initiative, carry a larger White advantage than drawish openings, and drawish openings have far more draws at every rating.

H3 **Time control.** White's advantage depends on how much time the players have. In slower games the first-move initiative is used more accurately and fewer games are decided by the clock, so we expect h to grow from Bullet to Classical.

2. Data

Source. The Lichess Open Database (`https://database.lichess.org`), standard rated games for **[MonthList]**, released under the Creative Commons CC0 licence. Each month is one zstd-compressed PGN file of tens of GB. Every game has about 17 header tags. We use seven: `Result`, `WhiteElo`, `BlackElo`, `Event` (gives the speed), `TimeControl`, `Termination` and `Opening`.

Month	Games read	Games kept	Kept
[filled in by white_advantage]			
Total	**[GamesRead]**	**[GamesUsed]**	**[PctUsed]**

Cleaning. We drop games with no result (**[SkipNoResult]**), abandoned games (**[SkipAbandoned]**), games with a missing rating (**[SkipNoRating]**), and games where the two ratings differ by more than 100 points (**[SkipGap]**). The rating gap is corrected for inside the model anyway, but very lopsided games say little about the first move. Each game is put in a 100-point rating bracket $[a, a{+}100)$ by the *average* of the two ratings, and in a speed (UltraBullet to Correspondence) from the `Event` tag.

Openings. Lichess names the opening of every game, for example *King's Gambit Accepted: Bishop's Gambit*. We match those names as shown in Table 3. Inside the drawish openings we leave out their gambit side-lines, such as the Stafford Gambit in the Petrov, so that the drawish group really is quiet. The eight openings cover **[OpeningGames]** games (**[OpeningShare]** of all kept games). Everything else is "other".

Why these eight. The aggressive openings are White gambits: White gives up a pawn or more for fast development and attack (King's, Evans, Danish and Smith-Morra Gambits). The drawish

Table 3: Openings and the Lichess opening names they cover. A game counts if its name starts with one of the listed names and contains none of the excluded ones.

Opening	Group	Lichess name starts with	Excluding
[filled in by white_advantage]			

openings have a reputation for symmetrical, simplified positions: the Petrov and the Berlin (Black's choice), the Queen's Gambit Declined (Black's choice), and the French Exchange (White's choice to simplify).

3. Method

The full derivations, with worked examples, are in the separate document *Theory.pdf*.

3.1 White's advantage in rating points

Under the Elo model, a player rated d points above the opponent is expected to score $E(d) = 1/(1+10^{-d/400})$. We give White an extra h points:

$$E_i(h) = \frac{1}{1+10^{-(R_i^{\mathrm{W}}+h-R_i^{\mathrm{B}})/400}},$$

and choose the h that makes the expected scores add up to the actual scores ($S_i = 1, \frac{1}{2}, 0$ for a White win, draw or loss). This is the maximum-likelihood estimate. We find it with Newton's method, and its standard error comes from the curvature of the log-likelihood (the Fisher information). Reporting h in rating points makes groups comparable: a 1540 who beats a 1570 with White counts as stronger evidence than a win between two 1570s. Draws count as half a point. Draw and win rates are reported separately.

3.2 h by rating (H1)

We fit h separately in every 100-point rating bracket (by the average of the two ratings), for each speed, and read the shape from the chart of h against rating. We don't fit a single line through the brackets, because the shape is not close to a straight line, and a slope would hide it.

3.3 Aggressive vs drawish (H2)

Each opening is treated as its own set of games: we fit h by maximum likelihood using only the games played in that opening, which gives one h (with its CI) per opening. For the comparison we also fit one h for all four aggressive openings together and one for the four drawish ones, and test the difference with a z-test: the two groups are different games, so the variances add, $\mathrm{SE}(\hat{h}_A - \hat{h}_D) = \sqrt{\mathrm{SE}_A^2 + \mathrm{SE}_D^2}$. We do this for all speeds together and for each of Bullet, Blitz and Rapid.

3.4 Time controls (H3)

Lichess sorts every game into a speed (UltraBullet, Bullet, Blitz, Rapid, Classical, Correspondence) by its time control. We treat each speed exactly like an opening: its games are a separate set, with their own maximum-likelihood h and win/draw/loss shares. We then compare each speed with the next faster one using the same z-test for a difference.

3.5 Big data and significance

With **[GamesUsed]** games, almost any difference has a tiny p-value. So we always report the size of an effect with its 95% confidence interval, and treat effects of a few rating points as small whatever

their p-value.

3.6 Tools

A C++ program streams the compressed files (zstd), reads only the header lines, and counts games by opening, speed, rating bracket and rating gap. The two months are read on two threads. We ran it on Google Colab, streaming the files straight from Lichess so nothing large was downloaded. It saves those counts (`counts.csv`, a few MB), and every table, test, figure, this report and an interactive Plotly dashboard (`output/final/dashboard.html`) are rebuilt from them in seconds.

4. Results

4.1 Overall

Over all kept games White wins **[WhiteAll]**, draws **[DrawAll]** and loses **[BlackAll]**. White's average score is **[ScoreAll]** against **[ExpectedAll]** predicted by the ratings alone, which is $h =$ **[HAll]** $\pm$ **[HAllCI]** rating points.

[figure FigResults, drawn by white_advantage]

Figure 5: Share of games won by White, drawn and won by Black, by rating (all speeds).

4.2 Does the advantage peak at intermediate ratings? (H1)

[interpretation: is H1 supported?]

[figure FigSpeed, drawn by white_advantage]

Figure 6: h in each 100-point rating bracket, by speed, with 95% CIs. Brackets with fewer than 1,000 games are left out.

4.3 Openings (H2)

Over all speeds, h for the aggressive openings minus h for the drawish openings is **[CompareDiff]** rating points ($z =$ **[CompareZ]**, $p =$ **[CompareP]**). **[interpretation: is H2 supported, and what does it say about White's advantage?]**

Table 4: White's advantage by opening (all speeds). Avg rating = average rating of the games.

Opening	Games	Avg rating	White	Draw	Black	h (pts)
[filled in by white_advantage]						
[filled in by white_advantage]						

[figure FigOpenings, drawn by white_advantage]

Figure 7: h for each opening (all speeds) with 95% CIs.

[figure FigGroupH, drawn by white_advantage]

Figure 8: h by rating for the aggressive and the drawish openings, with 95% CIs.

[figure FigGroupDraws, drawn by white_advantage]

Figure 9: Draw rate by rating for the aggressive and the drawish openings.

Table 5: Aggressive vs drawish openings: h for each group, and the difference with its z-test (rating points, 95% CI), for all speeds and for each of the main speeds.

Speed	Aggressive	Drawish	Difference	z	p
[filled in by white_advantage]					

4.4 Time controls (H3)

Table 6: White's advantage by speed (all openings). White / Draw / Black = share of games won by White, drawn, won by Black. h with 95% CI.

Speed	Games	Avg rating	White	Draw	Black	h (pts)
[filled in by white_advantage]						

[figure FigSpeedBars, drawn by white_advantage]

Figure 10: h for each speed (all openings) with 95% CIs.

Table 7: Each speed against the next faster one: h for both, and the difference (slower minus faster) with its z-test (rating points, 95% CI).

Speeds	Slower	Faster	Difference	z	p
[filled in by white_advantage]					

[figure FigSpeedDraws, drawn by white_advantage]

Figure 11: Draw rate by rating for each speed.

[interpretation: is H3 supported?]

5. Discussion

[what the results mean for White's advantage, written once the real results are in]

Limitations.

- *Choosing an opening is not random.* Players who play the King's Gambit or the Berlin differ in style and preparation, and the opening is chosen by one player (White for the gambits and the French Exchange, Black for the Petrov, Berlin and QGD). Our h corrects for the rating gap in each game, but the results show association, not that the opening causes the difference.
- *The groups have different players.* The comparison of the two opening groups doesn't adjust for their players' ratings. The "Avg rating" column and Figure 8 (which compares the groups bracket by bracket) show how much that matters.

- *Ratings are noisy*, especially for new accounts, which blurs the correction for the rating gap. Lichess ratings also differ from FIDE ratings, so "2200 on Lichess" is not a master.
- *Games are not independent.* Active players contribute thousands of games, so the real uncertainty is somewhat larger than our confidence intervals.
- *One site, two months.* The results describe Lichess players in early 2024.

6. Conclusion

[answer to the three questions in a few sentences]

Contributions

Sanidhya	Statistical modelling: fitting h, confidence intervals and the z-tests.
Daksh Maahor	Data cleaning and opening classification; win/draw/loss rates.
Sahil	Data engineering: the C++ streaming parser and the Colab pipeline.
VadaPavJai	Visualisation: figures, the Plotly dashboard and the report.

References

Lichess Open Database, `https://database.lichess.org`. Lichess opening names, `https://github.com/lichess-org/chess-openings`. A. E. Elo, *The Rating of Chessplayers, Past and Present*, Arco, 1978. M. E. Glickman, "Example of the Glicko-2 system", 2012.